

ADI Internship

Analog Basics (BGR & LDO)

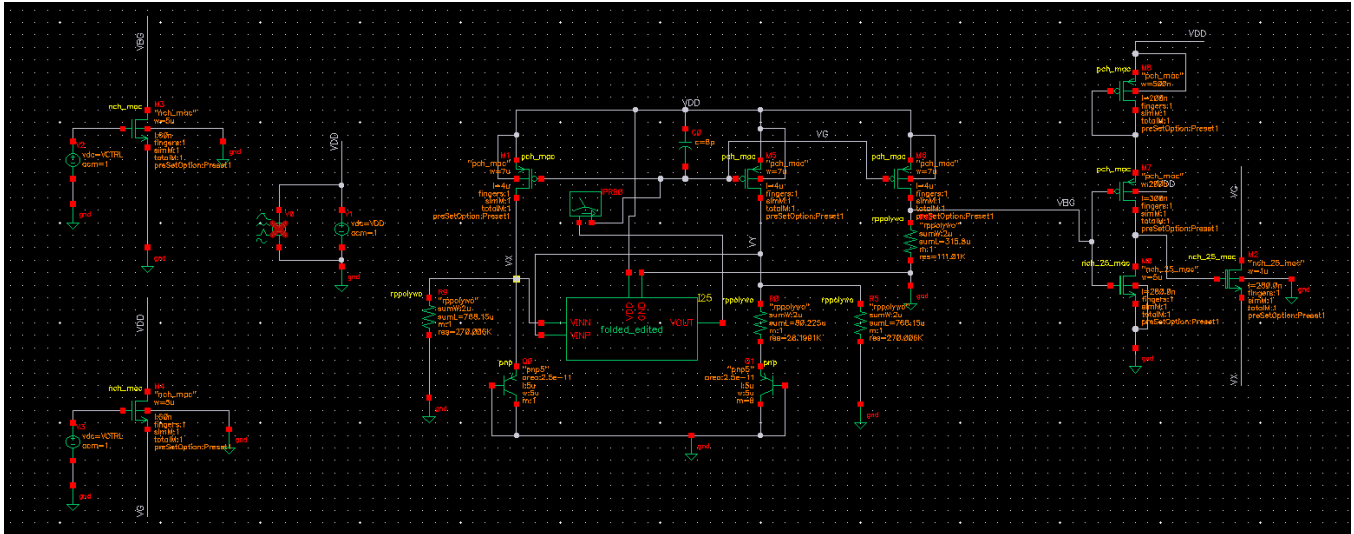
Prepared by
Kareem Ayman Saeed

Analog Basics: BGR & LDO

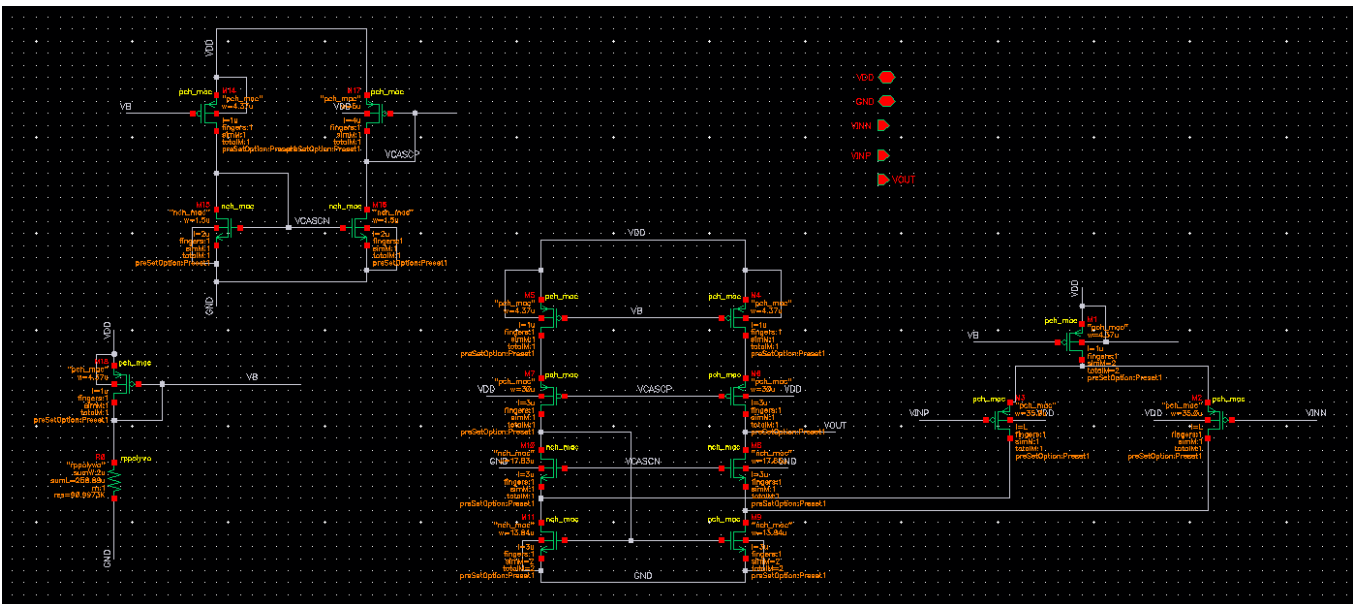
Part 1: BGR

1.Schematic

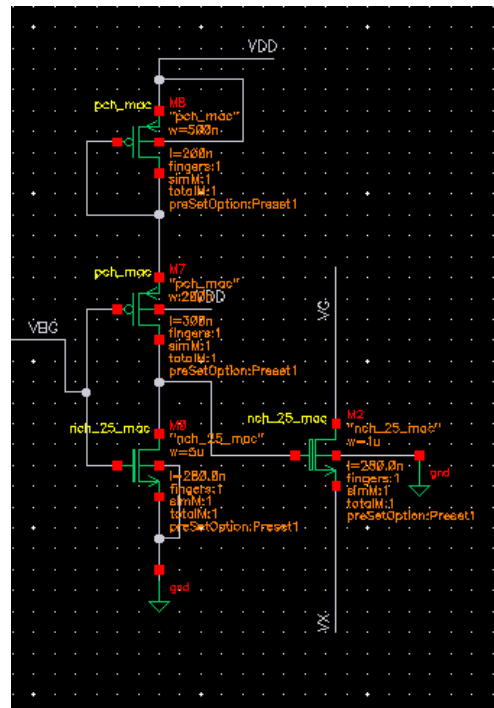
1.1 BGR



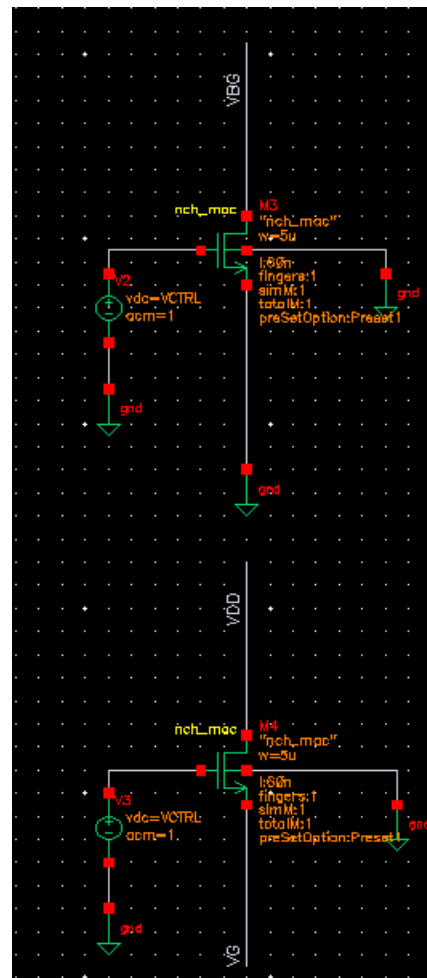
1.2 Op Amp (Folded Cascode)



1.3 Startup Circuit



1.4 Power Down Circuit



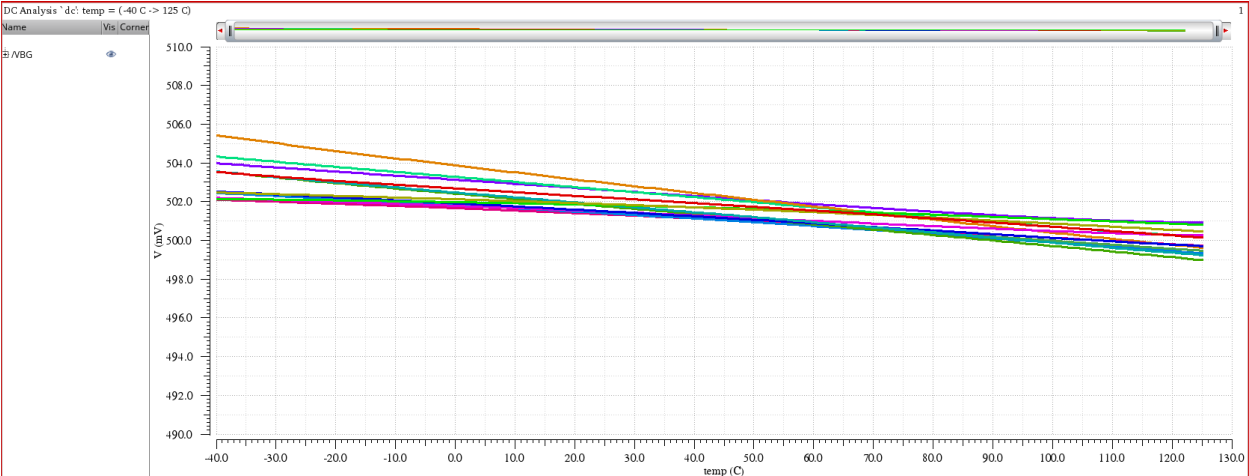
2. Results

2.1 V_{ref} Bandgap (Across PVT)

Parameter	Nominal					ff_0	ff_1	ff_2	fs_0	fs_1	fs_2	sf_0	sf_1	sf_2	ss_0	ss_1	ss_2
SaveOP.acs	1.5					1.35	1.5	1.65	1.35	1.5	1.65	1.35	1.5	1.65	1.35	1.5	1.65
VDD						ff_lib	ff_lib	ff_lib	fs_lib	fs_lib	fs_lib	sf_lib	sf_lib	sf_lib	ss_lib	ss_lib	ss_lib
toplevel.acs	tr_lib																

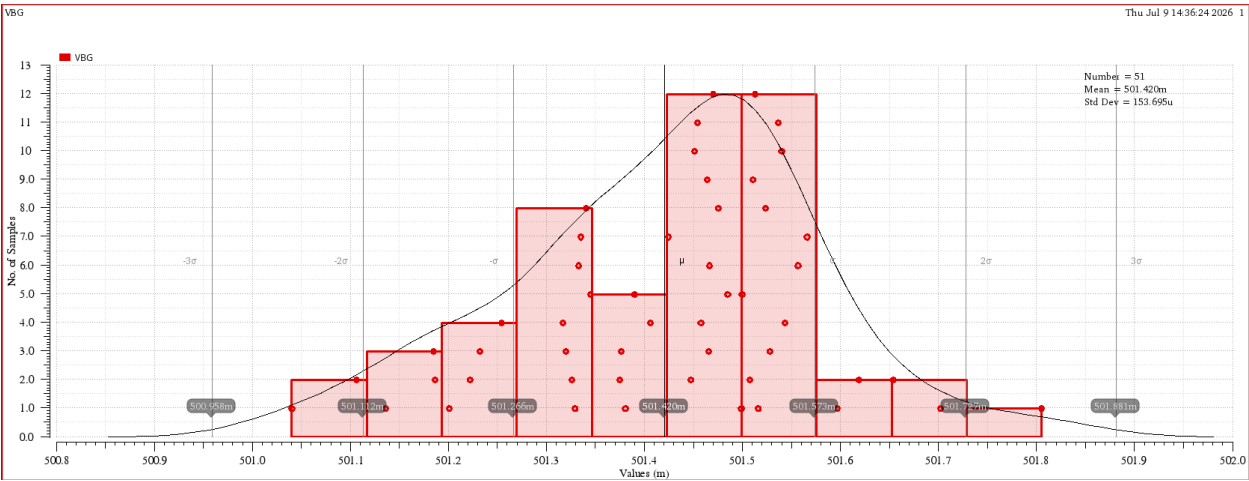
2 rows

Output	Nominal	Spec	Weight	Pass/Fail	Min	Max	ff_0	ff_1	ff_2	fs_0	fs_1	fs_2	sf_0	sf_1	sf_2	ss_0	ss_1	ss_2
BGR:1 ymax(v("VBG")?result"dc")	502.5m	< 510m		pass	502.1m	505.5m	505.5m	504.4m	504m	503.6m	502.5m	502.1m	503.6m	502.6m	502.2m	503.6m	502.5m	502.2m
BGR:1 ymin(v("VBG")?result"dc")	499.5m	> 490m		pass	499m	501m	499.7m	500.2m	501m	499m	499.4m	499.8m	499.3m	499.7m	500.3m	500.2m	500.5m	500.8m

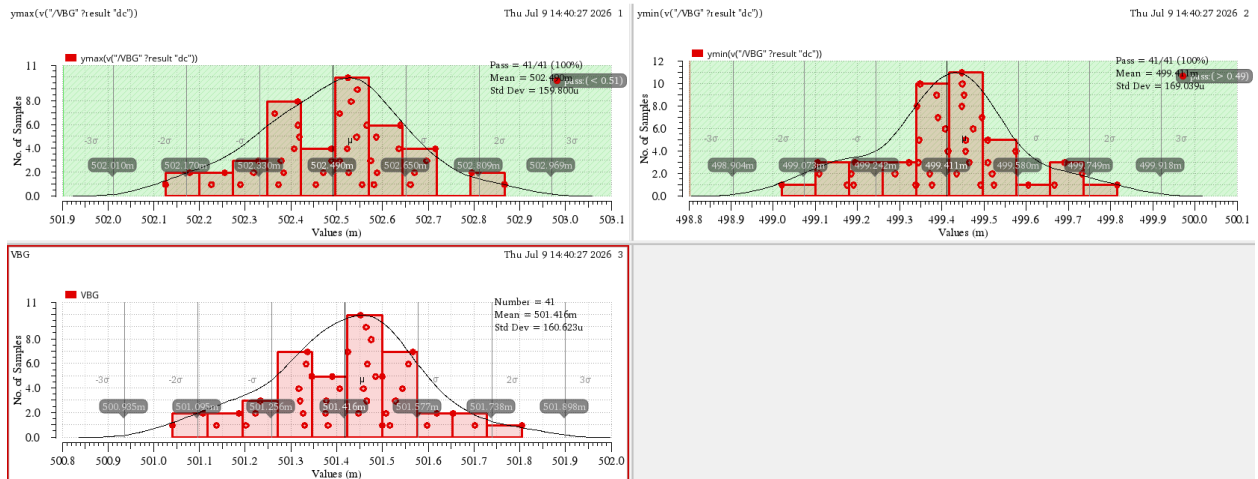


✓ Accuracy $\pm 2\%$ (across PVT)

2.2 V_{ref} Bandgap (Across PVT)



Test	Name	Yield	Min	Target	Max	Mean	Std Dev	Cpk	Errors
Yield Estimate: 20.5 % (41 passed/200 pts) Confidence Level: <not set> Filter: <not set>									
- Assignments: BGR:1									
-	ymax(v("VBG")?result"dc")	20.5	502.1m		502.9m	502.5m	159.8u	15.7	0
-	ymax(v("VBG")?result"dc")	20.5	502.1m	< 510m	502.9m	502.5m	159.8u	15.7	0
-	ymin(v("VBG")?result"dc")	20.5	499m		499.8m	499.4m	169u	18.6	0
-	ymin(v("VBG")?result"dc")	20.5	499m	> 490m	499.8m	499.4m	169u	18.6	0
-	VBG(su...)	20.5	501m		501.8m	501.4m	160.6u		0
-	VBG	20.5	501m	info	501.8m	501.4m	160.6u		0



✓ Accuracy $\pm 2\%$ (490mV: 510mV)

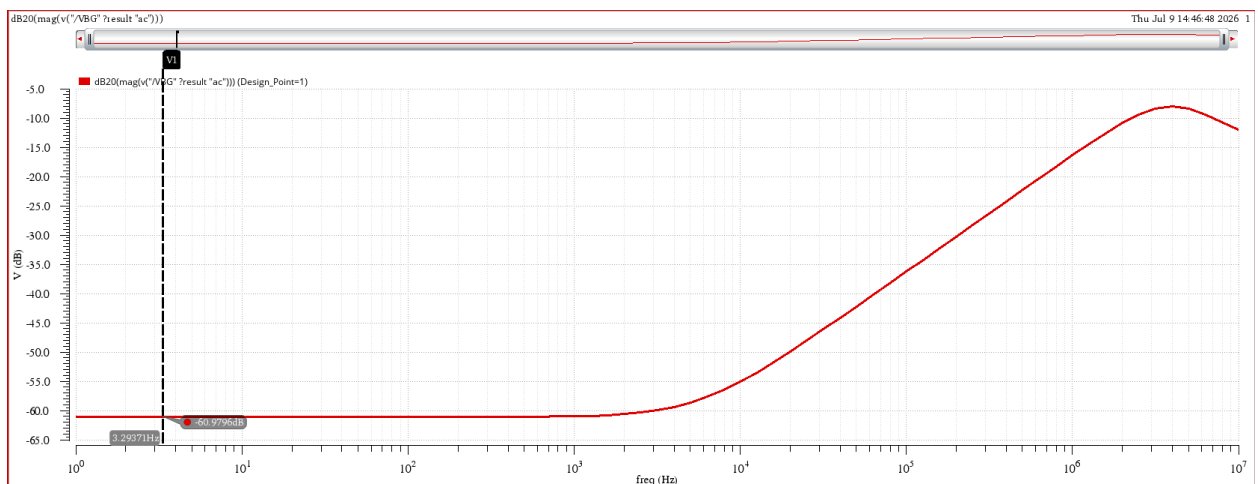
2.3 PSR (at DC)

Test	Output	Nominal	Spec	Weight	Pass/Fail
Filter	Filter	Filter	Filter	Filter	Filter
Assignments:BGR:1	dB20(mag(v('VBG'?result'ac')))				
Assignments:BGR:1	PSR	-60.98	< -60		pass

✓ Less than -60 dB

Parameter	Nominal	ff_0	ff_1	ff_2	fs_0	fs_1	fs_2	sf_0	sf_1	sf_2	ss_0	ss_1	ss_2
SaveOP.scs													
VDD	1.5	1.35	1.5	1.65	1.35	1.5	1.65	1.35	1.5	1.65	1.35	1.5	1.65
toplevel.scs	tt_lib	ff_lib	ff_lib	ff_lib	fs_lib	fs_lib	fs_lib	sf_lib	sf_lib	sf_lib	ss_lib	ss_lib	ss_lib

Test	Output	Nominal	Spec	Weight	Pass/Fail	Min	Max	ff_0	ff_1	ff_2	fs_0	fs_1	fs_2	sf_0	sf_1	sf_2	ss_0	ss_1	ss_2
Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
Assignments:BGR:1	PSR	-60.98	< -30		pass	-76.87	-47.4	-47.4	-62.13	-66.46	-47.56	-60.36	-76.87	-48.03	-61.91	-57.72	-48.1	-61.05	-74.93



✓ Less than -30 dB across PVT variations

2.4 PM & GM

Test	Output	Nominal	Spec	Weight	Pass/Fail
Filter	Filter	Filter	Filter	Filter	Filter
Assignments:BGR:1	PM	70.93	> 60		pass
Assignments:BGR:1	GM	20.51	> 10		pass

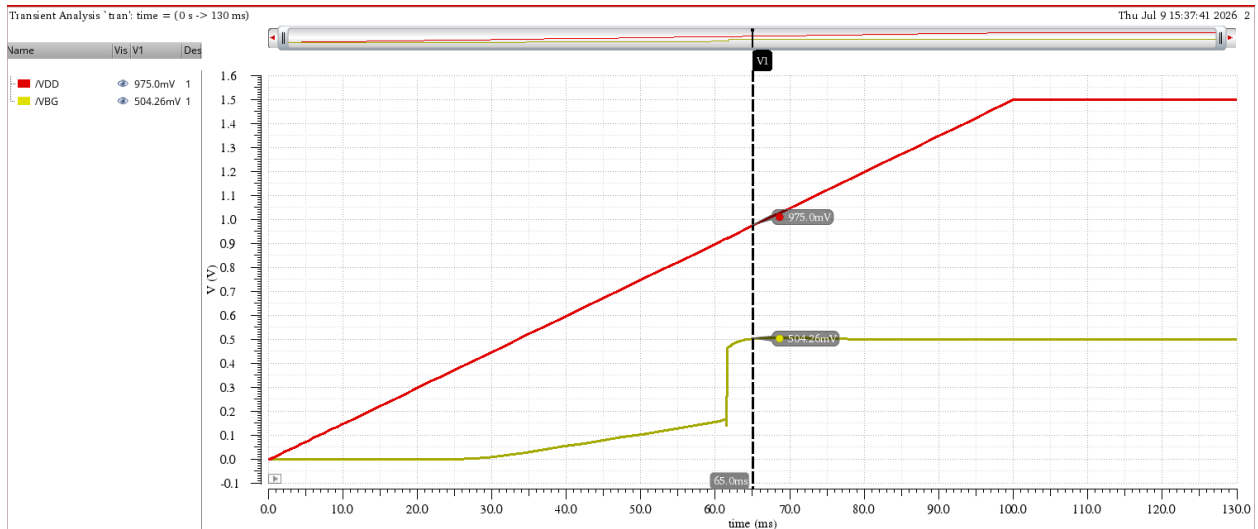
gainMargin(dB)=20.514822
 gainMarginFreq(Hz)=9831343.3
 phaseMargin(Deg)=70.913217
 phaseMarginFreq(Hz)=1638983.1

- ✓ PM greater than 60 & GM greater than 10 dB

Parameter	Nominal					ff_0	ff_1	ff_2	ff_3	ff_4	ff_5	ff_6	ff_7	ff_8	fs_0	fs_1	fs_2	fs_3	
SaveOP.scs																			
VDD	1.5					1.35	1.35	1.35	1.5	1.5	1.5	1.65	1.65	1.65	1.35	1.35	1.35	1.5	
temperature	27					-40	40	125	-40	40	125	-40	40	125	-40	40	125	-40	
toplevel.scs	tt_lib					ff_lib	ff_lib	ff_lib	ff_lib	ff_lib	ff_lib	ff_lib	ff_lib	ff_lib	fs_lib	fs_lib	fs_lib	fs_lib	
2 rows																			
Output	Nominal	Spec	Weight	Pass/Fail	Min	Max	ff_0	ff_1	ff_2	ff_3	ff_4	ff_5	ff_6	ff_7	ff_8	fs_0	fs_1	fs_2	fs_3
Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
PM	70.93	> 60		pass	61.63	77.97	70.06	74	77.97	65.14	72.74	77.14	63.73	71.68	76.47	73.3	73.81	77.59	64.84
GM	20.51	> 10		pass	17.05	23.48	18.05	21.07	23.48	17.87	21.4	23.31	18.59	21.29	23.11	19.08	20.78	23.34	17.44
fs_4	fs_5	fs_6	fs_7	fs_8	sf_0	sf_1	sf_2	sf_3	sf_4	sf_5	sf_6	sf_7	sf_8	ss_0	ss_1	ss_2	ss_3	ss_4	ss_5
Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
72.28	76.6	62.94	71.08	75.72	69.08	72.91	77.1	63.58	71.67	76.12	61.76	70.5	75.33	73.35	73	76.91	63.65	71.31	75.79
21.07	23.21	18.02	21.08	23.05	17.66	20.63	23.1	17.43	21.08	22.95	17.78	21.09	22.79	19.04	20.41	22.39	17.05	20.7	22.88
fs_7	fs_8	sf_0	sf_1	sf_2	sf_3	sf_4	sf_5	sf_6	sf_7	sf_8	ss_0	ss_1	ss_2	ss_3	ss_4	ss_5	ss_6	ss_7	ss_8
Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
71.08	75.72	69.08	72.91	77.1	63.58	71.67	76.12	61.76	70.5	75.33	73.35	73	76.91	63.65	71.31	75.79	61.63	70.35	74.81
21.08	23.05	17.66	20.63	23.1	17.43	21.08	22.95	17.78	21.09	22.79	19.04	20.41	22.39	17.05	20.7	22.88	17.59	21.02	22.7

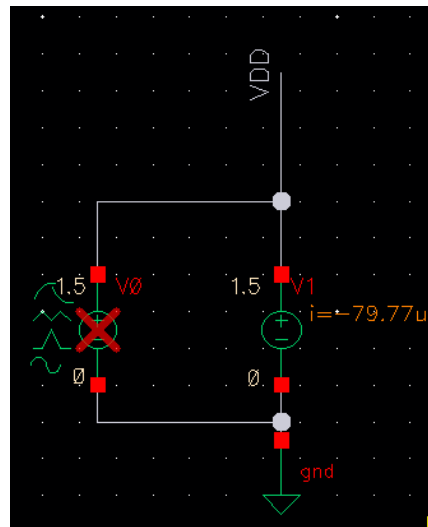
- ✓ PM greater than 60 & GM greater than 10 dB (across PVT)

2.5 Startup Speed



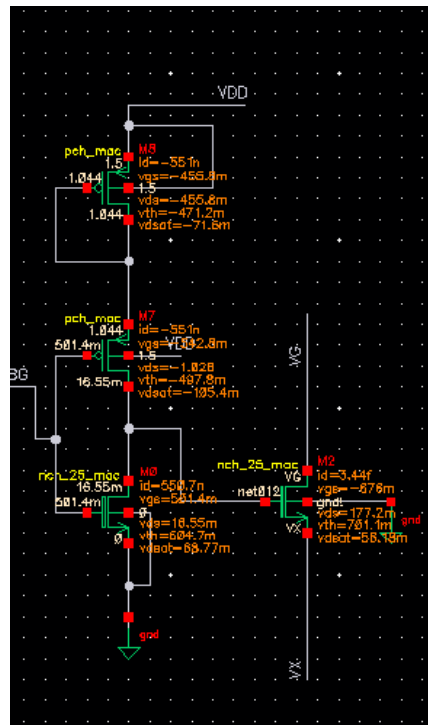
- ✓ Less than 80 msec (obtained: 65msec)

2.6 Current Consumption



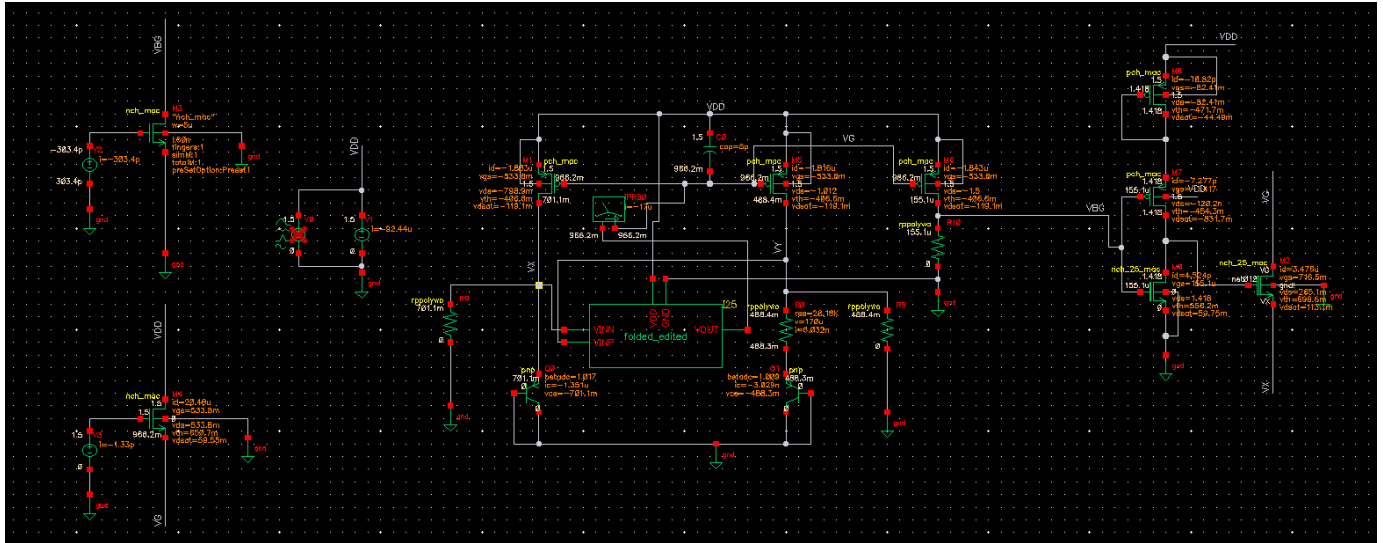
✓ Current Consumption: 79.77 μA (Including all circuits)

2.7 Startup Circuit Current when BG is on



✓ Startup Circuit Current when BG is on: 551 nA

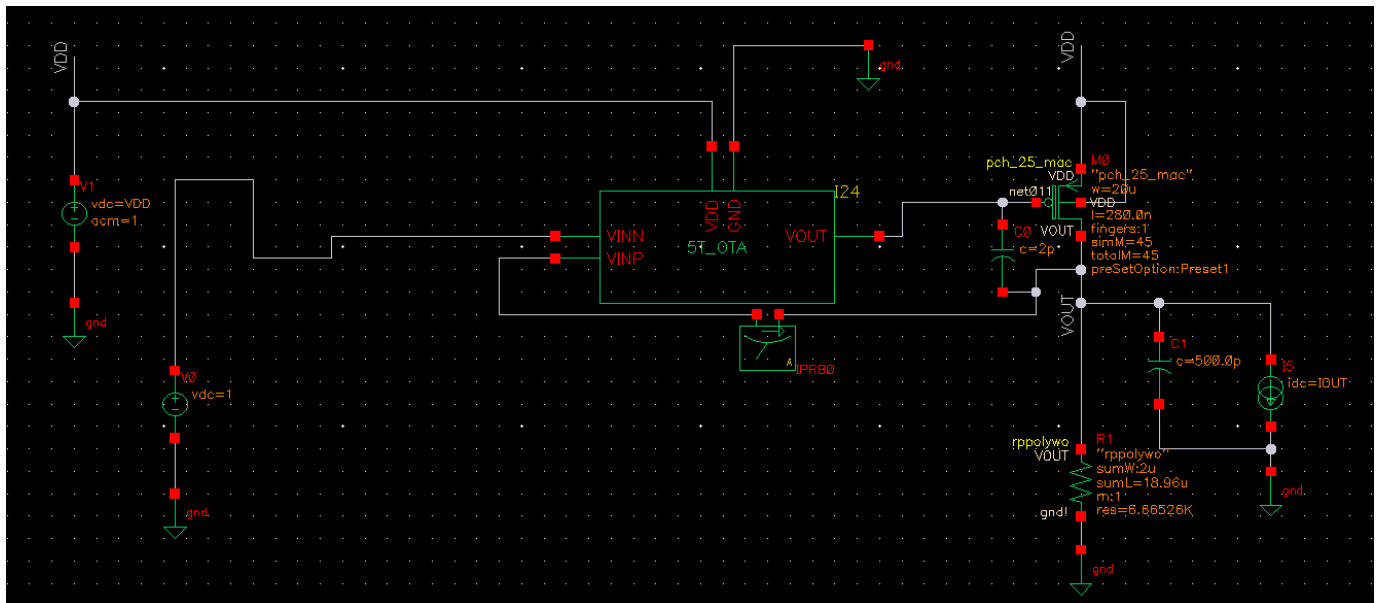
2.8 Power Down Current



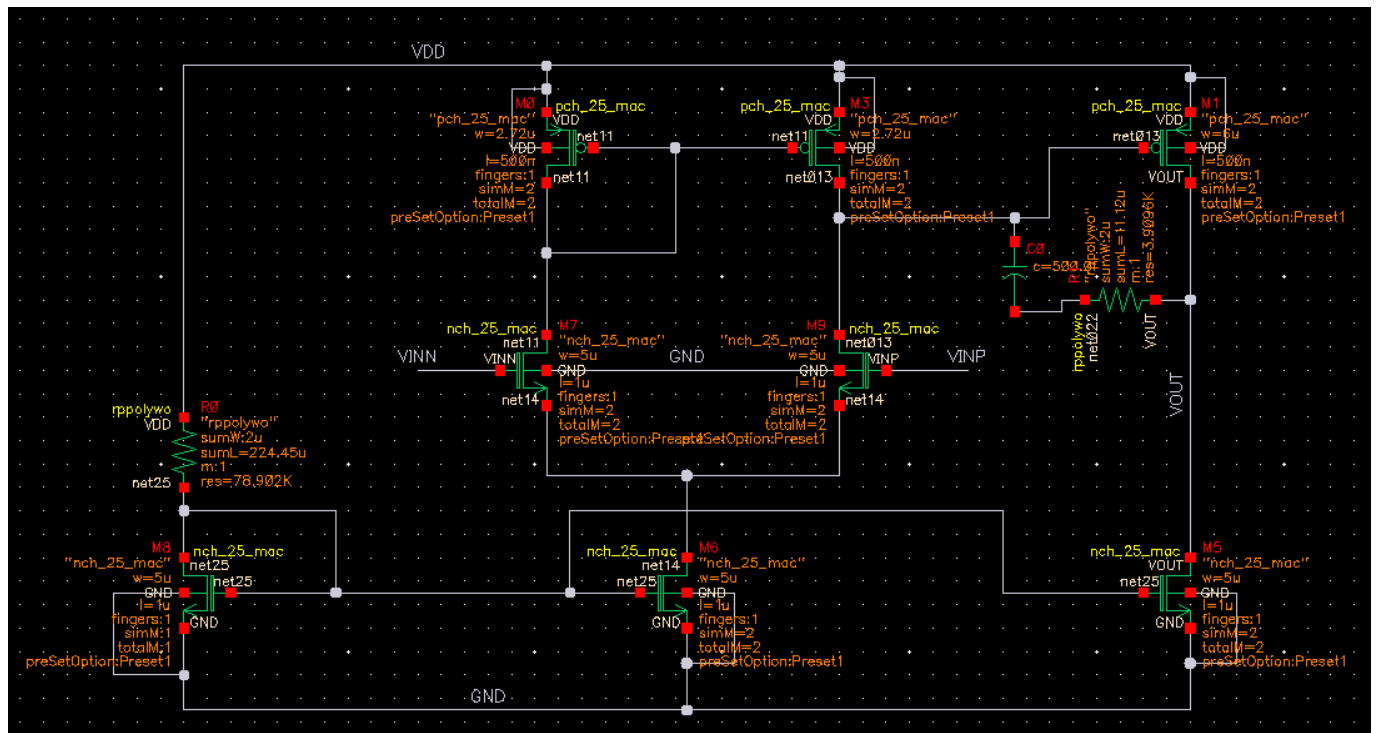
✓ Power Down Current: 92.44 μA (Including all circuits)

1.Schematic

1.1 LDO



1.2 Op Amp (Miller OTA)



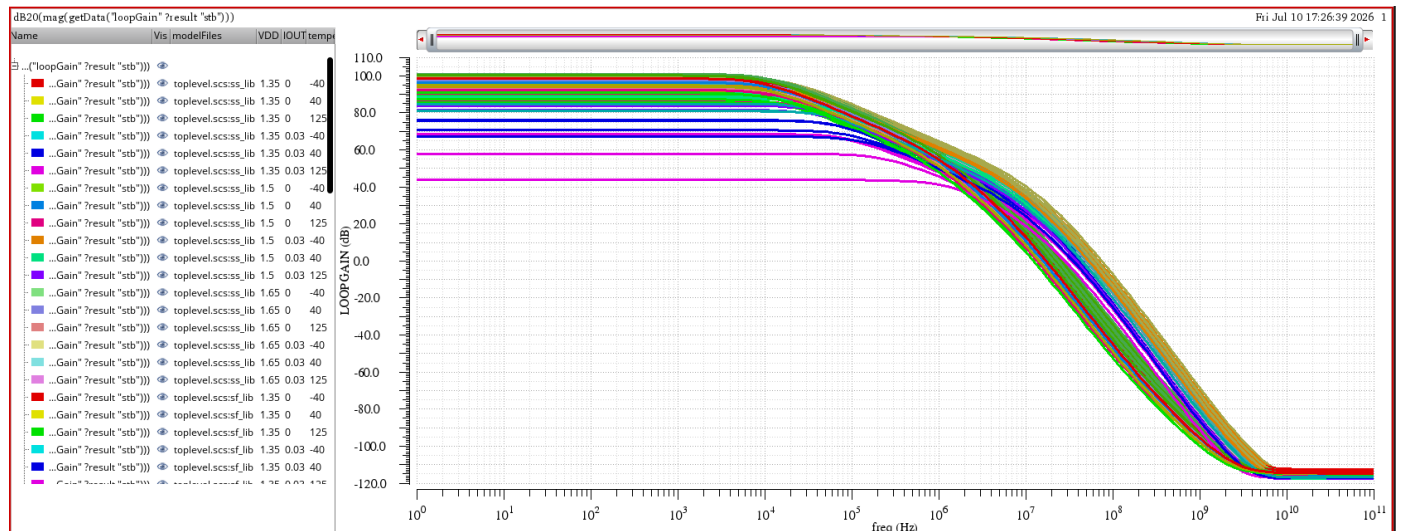
2. Results

2.1 GM & PM

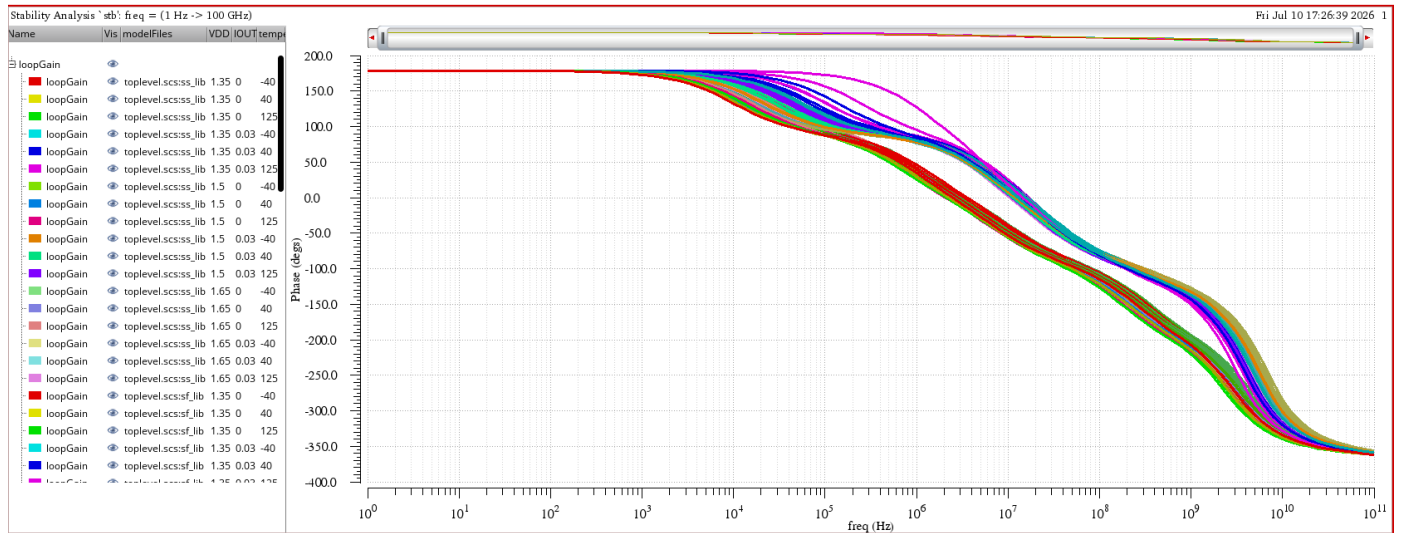
		Parameter	Nominal						ff_0	ff_1	ff_2	ff_3	ff_4	ff_5	ff_6	ff_7	ff_8	ss_0	ss_1
		VDD	1.5						1.35	1.35	1.35	1.5	1.5	1.5	1.65	1.65	1.65	1.35	1.35
		temperature	27						-40	40	125	-40	40	125	-40	40	125	-40	40
		toplevel.scs	tt_lib						ff_lib	ff_lib	ff_lib	ff_lib	ff_lib	ff_lib	ff_lib	ff_lib	ff_lib	ss_lib	ss_lib
5 rows																			
Point	Test	Output	Nominal	Spec	Weight	Pass/Fail	Min	Max	ff_0	ff_1	ff_2	ff_3	ff_4	ff_5	ff_6	ff_7	ff_8	ss_0	ss_1
Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
Parameters: IOUT=0																			
1	Assignments:LDO:1	PM	112	> 60		pass	107.2	125	113	116.9	122	110.3	113.9	118.3	109.3	112.4	116.3	110.8	114.1
1	Assignments:LDO:1	GM	38	> 10		pass	37.79	38.29	38.1	38.02	37.95	37.98	37.91	37.87	37.89	37.85	37.82	38.29	38.18
Parameters: IOUT=30m																			
2	Assignments:LDO:1	PM	120.5	> 60		pass	115.9	148.3	124.3	130.4	137.6	118.4	120.3	122.7	115.9	117.1	119.2	128.6	136.4
2	Assignments:LDO:1	GM	40.03	> 10		pass	39.84	40.21	39.84	39.89	39.94	39.89	39.91	39.96	39.89	39.95	39.98	40.09	40.17
5 rows																			
ss_2	ss_3	ss_4	ss_5	ss_6	ss_7	ss_8	sf_0	sf_1	sf_2	sf_3	sf_4	sf_5	sf_6	sf_7	sf_8	fs_0	fs_1	fs_2	fs_3
1.35	1.5	1.5	1.5	1.65	1.65	1.65	1.35	1.35	1.35	1.5	1.5	1.5	1.65	1.65	1.65	1.35	1.35	1.35	1.5
125	-40	40	125	-40	40	125	-40	40	125	-40	40	125	-40	40	125	-40	40	125	-40
ss_lib	ss_lib	ss_lib	ss_lib	ss_lib	ss_lib	ss_lib	sf_lib	sf_lib	sf_lib	sf_lib	sf_lib	sf_lib	sf_lib	sf_lib	sf_lib	fs_lib	fs_lib	fs_lib	fs_lib
rows																			
Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
117.4	108.4	110.7	114.2	107.2	109.6	112.3	109.7	113	117	108.7	111.2	114	107.7	109.7	113	114	119.1	125	111
38.09	38.13	38.04	37.98	38.02	37.95	37.91	38.21	38.12	38.07	38.09	38.03	37.94	38	37.93	37.86	38.13	38.05	38	38.02
148.3	119.9	121.9	125	117.1	117.6	119.5	125.9	132.6	140.1	119	120.5	124.1	116.6	117.9	118.5	126.7	134.4	143.5	118.9
40.17	40.09	40.16	40.19	40.13	40.17	40.21	39.99	40.03	40.05	39.98	40.05	40.07	40.02	40.05	40.13	39.97	40	40	39.97
5 rows																			
sf_7	sf_8	fs_0	fs_1	fs_2	fs_3	fs_4	fs_5	fs_6	fs_7	fs_8									
1.65	1.65	1.35	1.35	1.35	1.5	1.5	1.5	1.65	1.65	1.65									
40	125	-40	40	125	-40	40	125	-40	40	125	-40	40	125	-40	40	125	-40	40	125
sf_lib	sf_lib	fs_lib	fs_lib	fs_lib	fs_lib	fs_lib	fs_lib	fs_lib	fs_lib	fs_lib									
5 rows																			
Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
109.7	113	114	119.1	125	111	114.4	117.9	109	111.6	115.4									
37.93	37.86	38.13	38.05	38	38.02	37.96	37.88	37.93	37.86	37.79									
117.9	118.5	126.7	134.4	143.5	118.9	121	124.8	116.4	118	118.8									
40.05	40.13	39.97	40	40	39.97	40.04	40.04	40.01	40.03	40.1									

✓ PM greater than 60 & GM greater than 10 dB (across PVT)

- Loop Gain (Magnitude in dB)



- Loop Gain (Phase)



2.1 PSR

Parameter	Nominal	ff_0	ff_1	ff_2	ff_3	ff_4	ff_5	ff_6	ff_7	ff_8	ss_0
VDD	1.5	1.35	1.35	1.35	1.5	1.5	1.5	1.65	1.65	1.65	1.35
temperature	27	-40	40	125	-40	40	125	-40	40	125	-40
toplevel.scs	ss_lib	ss_lib	ss_lib	ss_lib	ss_lib	ss_lib	ss_lib	ss_lib	ss_lib	ss_lib	ss_lib

2 rows

Point	Test	Output	Nominal	Spec	Weight	Pass/Fail	Min	Max	ff_0	ff_1	ff_2	ff_3	ff_4	ff_5	ff_6	ff_7	ff_8	ss_0
1	Assignments:LDO:1	PSR	-97.13	<-30		pass	-122.1	-72.23	-83.18	-86.06	-106.4	-94.12	-122.1	-86.61	-111.7	-92.82	-83.26	-91.77
2	Assignments:LDO:1	PSR	-79.45	<-30		pass	-104	-39.03	-76.5	-69.72	-62.01	-89.4	-84.81	-73.54	-90.56	-86.48	-78.29	-72.17

Parameters: IOUT=0
Parameters: IOUT=30m

ss_1	ss_2	ss_3	ss_4	ss_5	ss_6	ss_7	ss_8	sf_0	sf_1	sf_2	sf_3	sf_4	sf_5	sf_6	sf_7	sf_8	fs_0	fs_1	fs_2
1.35	1.35	1.5	1.5	1.5	1.65	1.65	1.65	1.35	1.35	1.35	1.5	1.5	1.5	1.65	1.65	1.65	1.35	1.35	1.35
40	125	-40	40	125	-40	40	125	-40	40	125	-40	40	125	-40	40	125	-40	40	125
ss_lib	ss_lib	ss_lib	ss_lib	ss_lib	ss_lib	ss_lib	ss_lib	ss_lib	ss_lib	ss_lib	ss_lib	ss_lib	ss_lib	ss_lib	ss_lib	ss_lib	ss_lib	ss_lib	ss_lib

2 rows

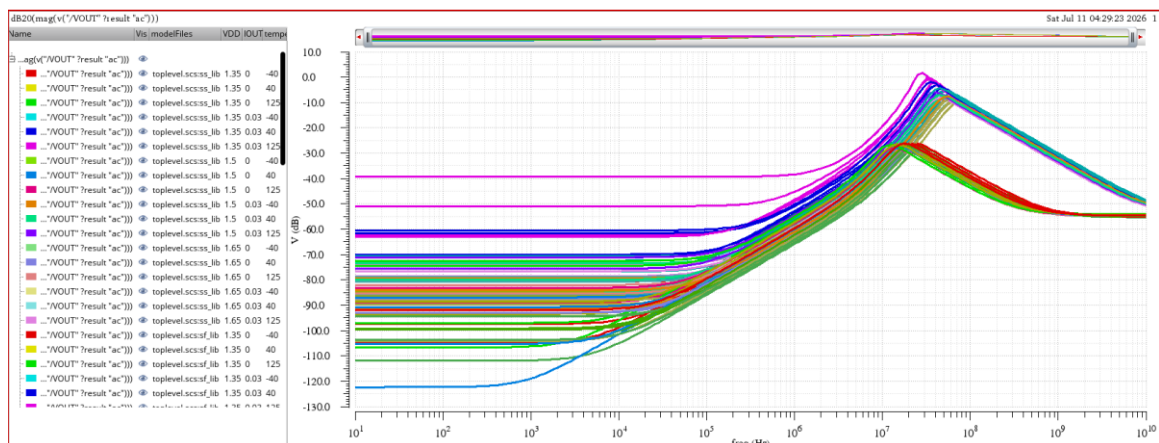
ss_1	ss_2	ss_3	ss_4	ss_5	ss_6	ss_7	ss_8	sf_0	sf_1	sf_2	sf_3	sf_4	sf_5	sf_6	sf_7	sf_8	fs_0	fs_1	fs_2
-88.49	-72.23	-98.91	-90.46	-82.9	-94.05	-92.7	-83.67	-80.45	-93.22	-96.76	-99.43	-105.1	-88.68	-103.4	-92.74	-85.14	-104.8	-84.27	-74.2
-60.23	-39.03	-83.82	-79.73	-70.83	-87.82	-80.43	-76.27	-80.5	-70.87	-62.75	-91.11	-83.38	-75.32	-104	-85.41	-80.37	-70.34	-61.32	-50.65

fs_0	fs_1	fs_2	fs_3	fs_4	fs_5	fs_6	fs_7	fs_8
1.35	1.35	1.35	1.5	1.5	1.5	1.65	1.65	1.65
-40	40	125	-40	40	125	-40	40	125
fs_lib	fs_lib	fs_lib	fs_lib	fs_lib	fs_lib	fs_lib	fs_lib	fs_lib

fs_0	fs_1	fs_2	fs_3	fs_4	fs_5	fs_6	fs_7	fs_8
-104.8	-84.27	-74.2	-103.6	-86.64	-79.89	-94.24	-87.32	-81.81
-70.34	-61.32	-50.65	-79.1	-73.16	-70.46	-89.56	-78.62	-73.97

✓ Less than -30 dB across PVT variations

- PSR (Magnitude in dB)

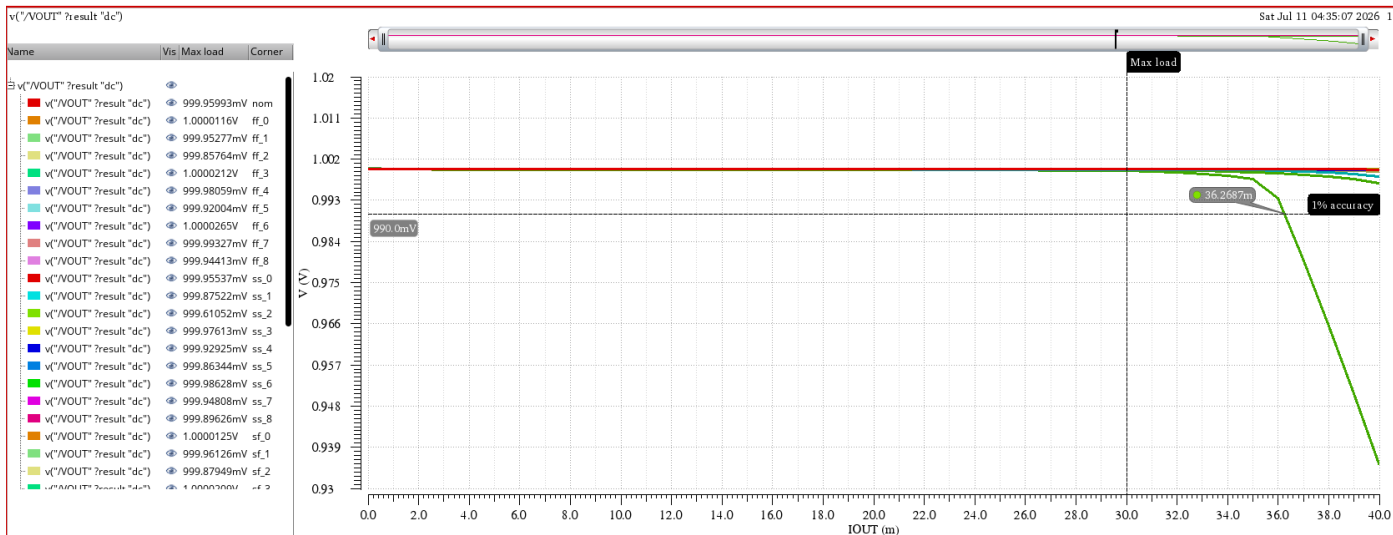


2.3 DC

		Parameter	Nominal						ff_0	ff_1	ff_2	ff_3	ff_4	ff_5	ff_6	ff_7	ff_8	ss_0	ss_1													
		VDD	1.5						1.35	1.35	1.35	1.5	1.5	1.5	1.65	1.65	1.65	1.35	1.35													
		temperature	27						-40	40	125	-40	40	125	-40	40	125	-40	40													
		toplevel.scs	tt_lib						ff_lib	ff_lib	ff_lib	ff_lib	ff_lib	ff_lib	ff_lib	ff_lib	ff_lib	ss_lib	ss_lib													
2 rows																																
Point	Filter	Test	Output	Nominal	Spec	Weight	Pass/Fail	Filter	Min	Max	Filter	ff_0	Filter	ff_1	Filter	ff_2	Filter	ff_3	Filter	ff_4	Filter	ff_5	Filter	ff_6	Filter	ff_7	Filter	ff_8	Filter	ss_0	Filter	ss_1
Parameters: IOUT=0																																
1		Assignments:LDO:1	VOUT	1	range 0.99 1.01		pass		999.8m	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	
Parameters: IOUT=30m																																
2		Assignments:LDO:1	VOUT	1	range 0.99 1.01		pass		999.6m	1	1	1	1	999.9m	1	1	999.9m	1	1	999.9m	1	1	999.9m	1	1	999.9m	1	1	999.9m	1	999.9m	
2 rows																																
ss_2	ss_3	ss_4	ss_5	ss_6	ss_7	ss_8	sf_0	sf_1	sf_2	sf_3	sf_4	sf_5	sf_6	sf_7	sf_8	fs_0	fs_1	fs_2	fs_3													
1.35	1.5	1.5	1.5	1.65	1.65	1.65	1.35	1.35	1.35	1.5	1.5	1.5	1.65	1.65	1.65	1.35	1.35	1.35	1.5													
125	-40	40	125	-40	40	125	-40	40	125	-40	40	125	-40	40	125	-40	40	125	-40													
ss_lib	ss_lib	ss_lib	ss_lib	ss_lib	ss_lib	ss_lib	sf_lib	sf_lib	sf_lib	sf_lib	sf_lib	sf_lib	sf_lib	sf_lib	sf_lib	sf_lib	fs_lib	fs_lib	fs_lib	fs_lib												
2 rows																																
Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter													
999.8m	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1													
h 999.6m	1	1	999.9m	1	999.9m	999.9m	1	1	999.9m	1	1	999.9m	1	1	1	999.9m	999.9m	999.7m	1													
2 rows																																
fs_1	fs_2	fs_3	fs_4	fs_5	fs_6	fs_7	fs_8																									
1.35	1.35	1.5	1.5	1.5	1.65	1.65	1.65																									
40	125	-40	40	125	-40	40	125																									
fs_lib	fs_lib	fs_lib	fs_lib	fs_lib	fs_lib	fs_lib	fs_lib																									
2 rows																																
fs_1	fs_2	fs_3	fs_4	fs_5	fs_6	fs_7	fs_8																									
Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter																									
1	1	1	1	1	1	1	1																									
999.9m	999.7m	1	999.9m	1	1	999.9m	999.9m																									

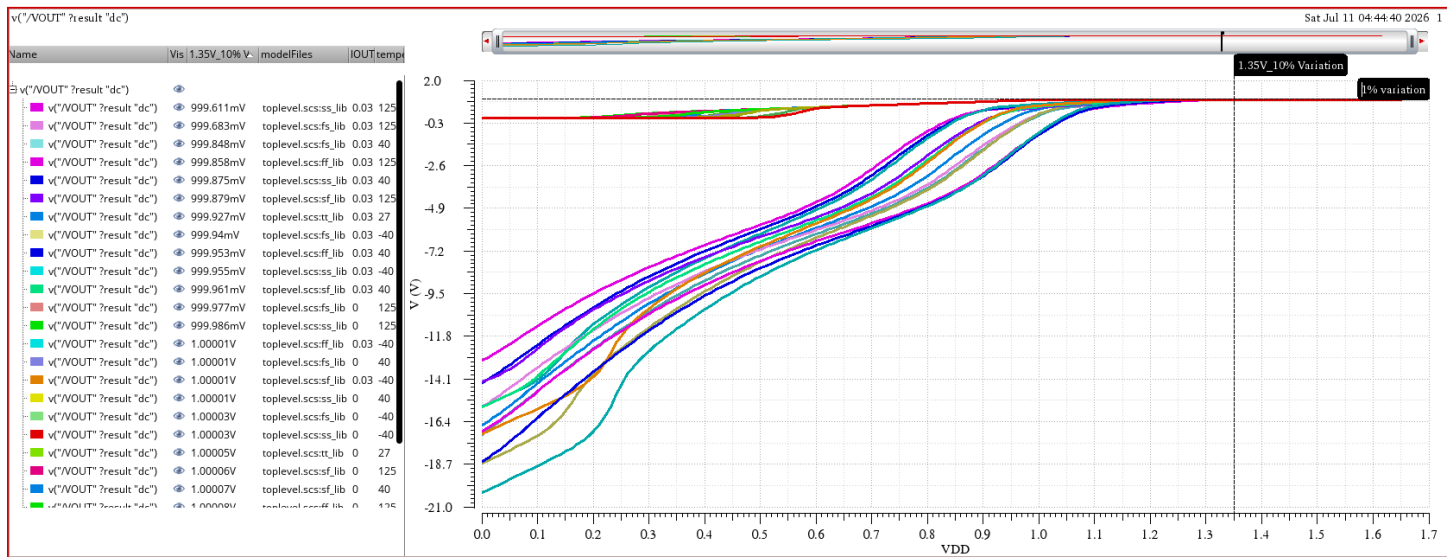
✓ Accuracy $\pm 1\%$ (990mV: 1.01V) across PVT

• Load Regulation



✓ Covers all loads below max load with Accuracy $\pm 1\%$ (990mV: 1.01V) across PVT

- Line Regulation ($I_{min} = 0A$ & $I_{max} = 30mA$)



✓ Covers line variation with Accuracy $\pm 1\%$ (990mV: 1.01V) across PVT